

Question 1-Sol.

(a) If $\underline{y} = [y_1, y_2, \dots, y_m]^T$ and $\underline{\theta} = [\theta_1, \theta_2, \dots, \theta_n]^T$, then model function $f(x; \underline{\theta})$ can be written as;

$$f(x; \underline{\theta}) = \underline{\theta}^T \underline{g}(x)$$

where $\underline{g}(x) = \begin{bmatrix} g_1(x) \\ g_2(x) \\ \vdots \\ g_n(x) \end{bmatrix}$. If the summation in the cost function is considered as a matrix

multiplication, the cost function can be rewritten as:

$$J(\theta) = (\underline{y} - \underline{A} \underline{\theta})^T (\underline{y} - \underline{A} \underline{\theta}) = \|\underline{y} - \underline{A} \underline{\theta}\|^2$$

where $\underline{A}$ matrix can be defined as :

$$\underline{A} = \begin{bmatrix} g_1(x_1) & \dots & g_n(x_1) \\ g_1(x_2) & \dots & g_n(x_2) \\ \vdots & \ddots & \vdots \\ g_1(x_m) & \dots & g_n(x_m) \end{bmatrix}.$$

Then the cost function $J(\theta)$ can be minimized with respect to $\underline{\theta}$, by taking its derivative and setting the derivative to zero. $J(\theta)$ can be rewritten as follows;

$$\begin{aligned} J(\theta) &= (\underline{y} - \underline{A} \underline{\theta})^T (\underline{y} - \underline{A} \underline{\theta}) \\ &= (\underline{y}^T - (\underline{A} \underline{\theta})^T) (\underline{y} - \underline{A} \underline{\theta}) \\ &= \underline{y}^T \underline{y} - 2(\underline{A} \underline{\theta})^T \underline{y} + (\underline{A} \underline{\theta})^T (\underline{A} \underline{\theta}) \\ &= \underline{y}^T \underline{y} - 2(\underline{A} \underline{\theta})^T \underline{y} + \underline{\theta}^T \underline{A}^T \underline{A} \underline{\theta} \end{aligned}$$

then the derivative of $J(\theta)$ can be written as:

$$\frac{d(J(\theta))}{d(\theta)} = -2\underline{A}^T \underline{y} + 2\underline{A}^T \underline{A} \underline{\theta} = 0$$

then closed form solution for the optimal $\underline{\theta}^*$ can be found as:

$$\frac{d(J(\theta))}{d(\theta)} = -2\underline{A}^T \underline{y} + 2\underline{A}^T \underline{A} \underline{\theta} = 0 \Rightarrow \underline{\theta}^* = (\underline{A}^T \underline{A})^{-1} \underline{A}^T \underline{y}.$$

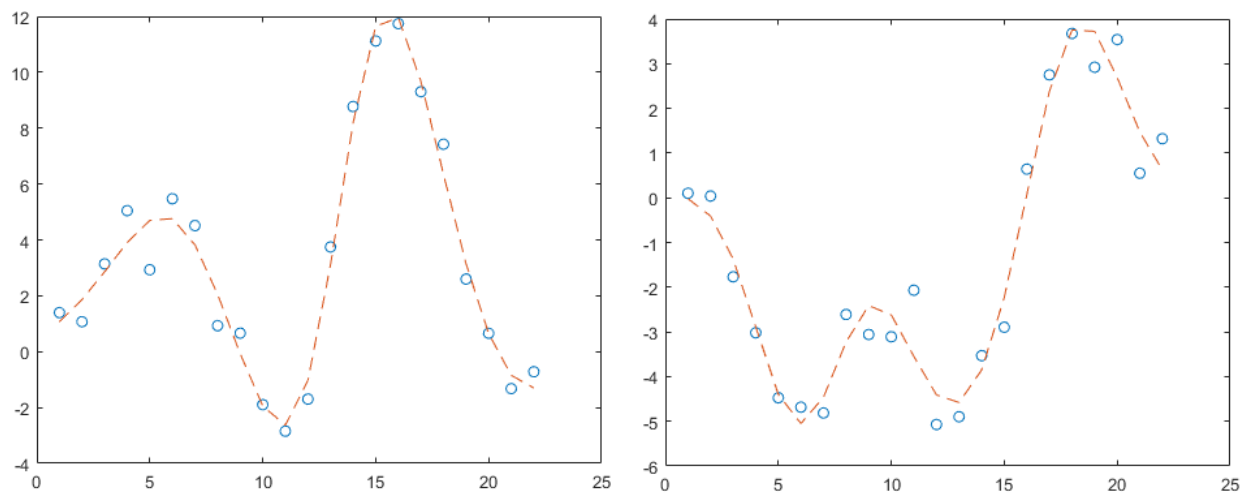
(b) An example implementation of MATLAB can be as follows:

```

1 clear all
2 close all
3 clc
4
5 m =22;
6 n = 7;
7 Mu = 3;
8 sigma = 2;
9
10 % Generate x and y pairs
11 x = [1:1:m]';
12 X = repmat(x,1,n);
13 A = hypothesis(X,Mu,sigma);
14
15 % Generate random noise and random theta parameters
16 theta = 5*randn(n,1);
17 noise = randn(m,1);
18
19 y = A*theta+noise;
20
21 % Obtain theta by using the result of the minimization
22 theta_opt = inv(A'*A)*A'*y;
23
24 % Obtain curve fitting result y-s
25 y_s = hypothesis(X,Mu,sigma)*theta_opt;
26
27 % Plot the curve fit
28 plot(x,y,'o',x,y_s,'--')
29
30 function out = hypothesis(x,Mu,sigma)
31 [m,n] = size(x);
32 MU_mult = 1:1:n;
33 MU_mult = Mu*MU_mult;
34 MU_mult = repmat(MU_mult,m,1);
35 out = exp(-(x-MU_mult).^2)./(2*sigma^2) );
36 end

```

2 example outputs of this MATLAB script are as follows:



Q2

(a) The condition for invertability is $\det(\cdot) \neq 0$ since,

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)} \quad \text{where } A \text{ is } n \times n \text{ matrix.}$$

For the entire family of $A + cI$ where c is a complex number
 Given, $c = a + bi$ where a, b are any constants and $i = \sqrt{-1}$

$$(A + cI)x = \lambda x$$

$$(A + cI - \lambda I)x = 0$$

$$(A + (c - \lambda)I)x = 0$$

If the eigenvalues of A is real without any complex part then
 the entire family of $A + cI$ is invertable

(b) Similarly, the eigenvalues of the matrix A must be complex since r
 is real. Also, the elements of A is to be real (i.e., real matrix
with complex eigenvalues). The one type of ~~are~~ this matrices are
 rotation matrices:

$$R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$\text{for } \theta = 30^\circ, R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.8660 & -0.5000 \\ 0 & 0.5000 & 0.8660 \end{bmatrix}$$

$$\lambda_{1,2} = 0.8660 \pm 0.5i$$

$$\lambda_3 = 1$$

Q3 If K is skew-symmetric, $K^T = -K$ or $K + K^T = 0$

$$Q = (I - K)(I + K)^{-1}$$

$$Q^T = [(I - K)(I + K)^{-1}]^T$$

$$= [(I + K)^{-1}]^T (I - K)^T$$

$$= [(I + K)^T]^{-1} (I - K)^T$$

$$= (I^T + K^T)^{-1} (I^T - K^T)$$

$$= (I + K^T)^{-1} (I - K^T)$$

$$= (I - K)^{-1} (I + K)$$

$$= (I + K)(I - K)^{-1}$$

$$(AB)^T = B^T A^T$$

$$(A^{-1})^T = (A^T)^{-1}$$

$$(A + B)^T = A^T + B^T$$

$$I^T = I$$

$$K^T = -K$$

Observe that $(I - K)(I + K) = I + K - K - K^2 = I - K^2$
 $(I + K)(I - K) = I - K + K - K^2 = I - K^2$

That is, $(I - K)(I + K) = (I + K)(I - K)$

$$\Rightarrow (I + K) = (I - K)^{-1} (I + K)(I - K)$$

$$\Rightarrow (I + K)(I - K)^{-1} = (I - K)^{-1} (I + K)$$

So, $Q = (I - K)(I + K)^{-1}$

$$Q^T = (I + K)(I - K)^{-1}$$

$$Q Q^T = (I - K) \underbrace{(I + K)^{-1} (I + K)}_I \underbrace{(I - K)^{-1}}_I = I$$

So Q is orthogonal.

for $K = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

$$Q = (I - K)(I + K)^{-1} = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1/5 & -2/5 \\ 2/5 & 1/5 \end{bmatrix}$$

$$\Rightarrow Q = \begin{bmatrix} -3/5 & -4/5 \\ 4/5 & -3/5 \end{bmatrix}$$

Q4 Using the fact that $A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = \det(A) \cdot I$

(Eq1) — $Ax = \lambda x$

↓ multiplying $\text{adj}(A)$ from the left

$$\text{adj}(A) \cdot A \cdot x = \text{adj}(A) \cdot \lambda \cdot x$$

$$\det(A)$$

$$\det(A) \cdot x = \lambda \cdot \text{adj}(A) \cdot x$$

↓ λ is a scalar and can be moved to the left

$$\Rightarrow \frac{\det(A)}{\lambda} \cdot x = \text{adj}(A) \cdot x \iff \text{adj}(A) \cdot x = \underbrace{\frac{\det(A)}{\lambda}}_{\text{eigenvalue}} \cdot x \quad \text{(Eq2)}$$

λ is a nonzero scalar

(a) since $\text{adj}(A) \cdot x = \frac{\det(A)}{\lambda} \cdot x$ and $x \neq 0$
(it is an eigenvector of A corresponding to λ), $\frac{\det(A)}{\lambda}$ is an eigenvalue of $\text{adj}(A)$.

(b) Since (Eq2) is derived from (Eq1), both has the same eigenvector.

Q5

Given,

$$Ax = \lambda_1 x$$

and

$$A^T y = \lambda_2 y$$

taking
transpose

$$(Ax)^T = (\lambda_1 x)^T$$

$$(AB)^T = B^T A^T$$

$$x^T A^T = \lambda_1 \cdot x^T$$

multiplying
y from the
right

$$x^T \cdot \underbrace{A^T \cdot y}_{\lambda_2 y} = \lambda_1 \cdot x^T \cdot y$$

λ_2 is
a scalar

$$\Rightarrow x^T \cdot \lambda_2 \cdot y = \lambda_1 \cdot x^T \cdot y$$

$$\lambda_2 \cdot x^T \cdot y = \lambda_1 \cdot x^T \cdot y$$

$$\Rightarrow x^T y (\lambda_1 - \lambda_2) = 0$$

since $\lambda_1 \neq \lambda_2$

$$x^T y = 0 //$$

Q6 First check whether A is diagonalizable or not;

$$\det(\lambda I - A) = \begin{vmatrix} \lambda-1 & -1 & 0 \\ 0 & \lambda-2 & 0 \\ 0 & -1 & \lambda-2 \end{vmatrix} = (\lambda-1)(\lambda-2)^2$$

$$\lambda_1 = 1$$

$$\lambda_{2,3} = 2$$

for λ_1 ;

$$(\lambda_1 I - A)v_1 = 0 = \begin{bmatrix} 0 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & -1 & -1 \end{bmatrix} v_1 = 0 \Rightarrow v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

for $\lambda_{2,3}$

$$(\lambda_{2,3} I - A)v_{2,3} = 0 \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} v_{2,3} = 0 \Rightarrow v_{2,3} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Since A has 2 L.I. eigenvectors, not 3, it is not diagonalizable.

Defining a polynomial with degree of $n-1 = 3-1 = 2$ as;

$$h(\lambda) := \beta_0 + \beta_1 \lambda + \beta_2 \lambda^2$$

where $f^{(l)}(\lambda_i) = h^{(l)}(\lambda_i)$ for $l = 0, 1, \dots, n_i - 1$ and $i = 1, 2, \dots, m$

also $f^{(l)}(\lambda_i) := \frac{d^l f(\lambda)}{d\lambda^l} \Big|_{\lambda=\lambda_i}$ and $h^{(l)}(\lambda_i) := \frac{d^l h(\lambda)}{d\lambda^l} \Big|_{\lambda=\lambda_i}$

so we have $f(A) = h(A)$ and $h(\lambda)$ is said to equal $f(\lambda)$ on the spectrum of A . We want to find $f(x) = e^x$ and $f(x) = e^{-x}$.

(Theorem 3.5 of 3rd ed. of Linear System Theory and Design by Chi-Tsong Chen, p. 64)

for $\lambda_1 = 1$; $\begin{cases} f(1) = h(1) \\ f'(1) = h'(1) \end{cases}$; $\begin{cases} e^1 = \beta_0 + \beta_1 + \beta_2 \\ e^2 = \beta_0 + 2\beta_1 + 4\beta_2 \end{cases}$; $\begin{cases} e^2 - e = \beta_1 + 3\beta_2 \\ e^2 - 4e = \beta_1 \end{cases}$; $\begin{cases} e = \beta_2 \\ 4e - e^2 = \beta_0 \end{cases}$

for $\lambda_{2,3} = 2$; $\begin{cases} f(2) = h(2) \\ f'(2) = h'(2) \end{cases}$; $\begin{cases} e^2 = \beta_0 + 2\beta_1 + 4\beta_2 \\ e^2 = \beta_1 + 4\beta_2 \end{cases}$; $\begin{cases} e^2 - e = \beta_1 + 3\beta_2 \\ e^2 - 4e = \beta_1 \end{cases}$; $\begin{cases} e = \beta_2 \\ 4e - e^2 = \beta_0 \end{cases}$

$$\Rightarrow h(\lambda) = (4e - e^2) + (e^2 - 4e)\lambda + e\lambda^2$$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\Rightarrow f(A) = (4e - e^2)I + (e^2 - 4e)A + eA^2$$

$$A^2 = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 4 & 0 \\ 0 & 4 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4e - e^2 & 0 & 0 \\ 0 & 4e - e^2 & 0 \\ 0 & 0 & 4e - e^2 \end{bmatrix} + \begin{bmatrix} e^2 - 4e & e^2 - 4e & 0 \\ 0 & 2e^2 - 8e & 0 \\ 0 & e^2 - 4e & 2e^2 - 8e \end{bmatrix}$$

$$+ \begin{bmatrix} e & 3e & 0 \\ 0 & 4e & 0 \\ 0 & 4e & 4e \end{bmatrix}$$

$$e^A = \begin{bmatrix} e & e^2 - e & 0 \\ 0 & e^2 & 0 \\ 0 & e^2 & e^2 \end{bmatrix}$$

for $f(x) = e^{-x}$

$$\begin{aligned} \lambda_1 = 1; \quad f(1) = h(1): \quad & e^{-1} = \beta_0 + \beta_1 + \beta_2 \\ \lambda_2 = 2; \quad f(2) = h(2): \quad & e^{-2} = \beta_0 + 2\beta_1 + 4\beta_2 \\ f'(2) = h'(2): \quad & -e^{-2} = \beta_1 + 4\beta_2 \end{aligned} \quad \left. \begin{aligned} & e^{-1} = \beta_0 + \beta_1 + \beta_2 \\ & e^{-2} = \beta_0 + 2\beta_1 + 4\beta_2 \\ & -e^{-2} = \beta_1 + 4\beta_2 \end{aligned} \right\} \begin{aligned} & e^{-2} - e^{-1} = \beta_1 + 3\beta_2 \\ & \underline{e^{-1} - 2e^{-2} = \beta_2} \\ & \underline{7e^{-2} - 4e^{-1} = \beta_1} \\ & \underline{4e^{-1} - 5e^{-2} = \beta_0} \end{aligned}$$

$$\Rightarrow h(\lambda) = (4e^{-1} - 5e^{-2}) + (7e^{-2} - 4e^{-1})\lambda + (e^{-1} - 2e^{-2})\lambda^2$$

$$\Rightarrow f(A) = (4e^{-1} - 5e^{-2})I + (7e^{-2} - 4e^{-1})A + (e^{-1} - 2e^{-2})A^2$$

$$= \begin{bmatrix} 4e^{-1} - 5e^{-2} & 0 & 0 \\ 0 & 4e^{-1} - 5e^{-2} & 0 \\ 0 & 0 & 4e^{-1} - 5e^{-2} \end{bmatrix} + \begin{bmatrix} 7e^{-2} - 4e^{-1} & 7e^{-2} - 4e^{-1} & 0 \\ 0 & 14e^{-2} - 8e^{-1} & 0 \\ 0 & 7e^{-2} - 4e^{-1} & 14e^{-2} - 8e^{-1} \end{bmatrix}$$

$$+ \begin{bmatrix} e^{-1} - 2e^{-2} & 3e^{-1} - 6e^{-2} & 0 \\ 0 & 4e^{-1} - 8e^{-2} & 0 \\ 0 & 4e^{-1} - 8e^{-2} & 4e^{-1} - 8e^{-2} \end{bmatrix}$$

$$e^{-A} = \begin{bmatrix} e^{-1} & e^{-2} - e^{-1} & 0 \\ 0 & e^{-2} & 0 \\ 0 & -e^{-2} & e^{-2} \end{bmatrix}$$

$$\frac{e^A - e^{-A}}{e^A - e^{-A}} = \underbrace{\begin{bmatrix} e+e^{-1} & e^2-e^{-2}-e+e^{-1} & 0 \\ 0 & e^2+e^{-2} & 0 \\ 0 & e^2-e^{-2} & e^2+e^{-2} \end{bmatrix}}_{e^A - e^{-A}} \underbrace{\begin{bmatrix} e-e^{-1} & e^2-e^{-2}-e+e^{-1} & 0 \\ 0 & e^2-e^{-2} & 0 \\ 0 & e^2+e^{-2} & e^2-e^{-2} \end{bmatrix}}_{(e^A - e^{-A})^{-1}}^{-1}$$

$$= \begin{bmatrix} e+e^{-1} & e^2-e^{-2}-e+e^{-1} & 0 \\ 0 & e^2+e^{-2} & 0 \\ 0 & e^2-e^{-2} & e^2+e^{-2} \end{bmatrix} \begin{bmatrix} 1 & \frac{-(e^2-e^{-2}-e+e^{-1})}{(e-e^{-1})(e^2-e^{-2})} & 0 \\ 0 & 0 & 0 \\ 0 & \frac{-(e^2+e^{-2})}{(e^2-e^{-2})(e^2-e^{-2})} & \frac{1}{e^2-e^{-2}} \end{bmatrix}$$

$$f(A) = \begin{bmatrix} e+e^{-1} & \frac{-(e+e^{-1})(e^2-e^{-2}-e+e^{-1})}{(e-e^{-1})(e^2-e^{-2})} & 0 \\ 0 & 0 & 0 \\ 0 & \frac{-(e^2+e^{-2})}{(e^2-e^{-2})} & \frac{e^2+e^{-2}}{e^2-e^{-2}} \end{bmatrix} //$$